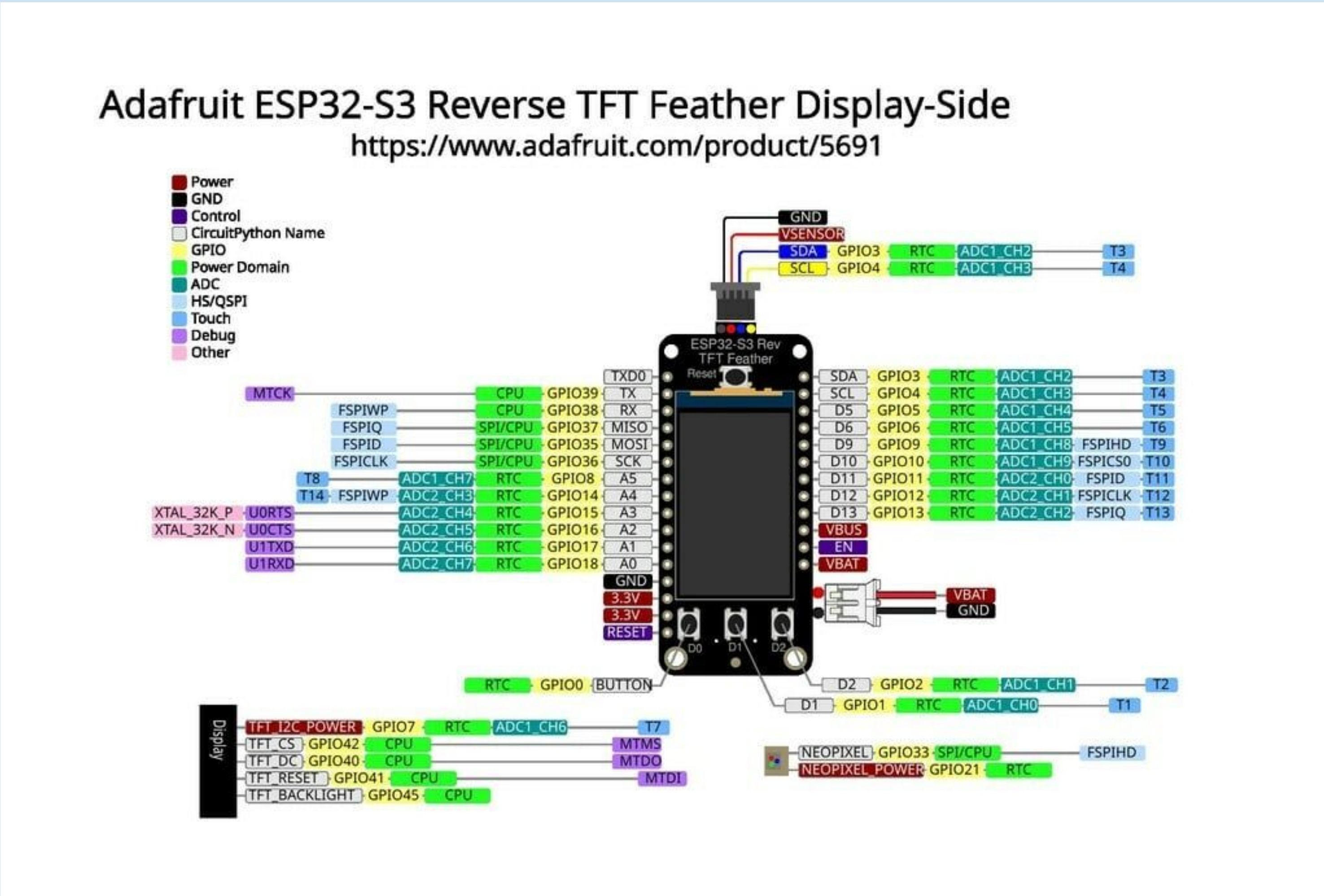
testingConnection Example

2026

Overview of your ESP32 board

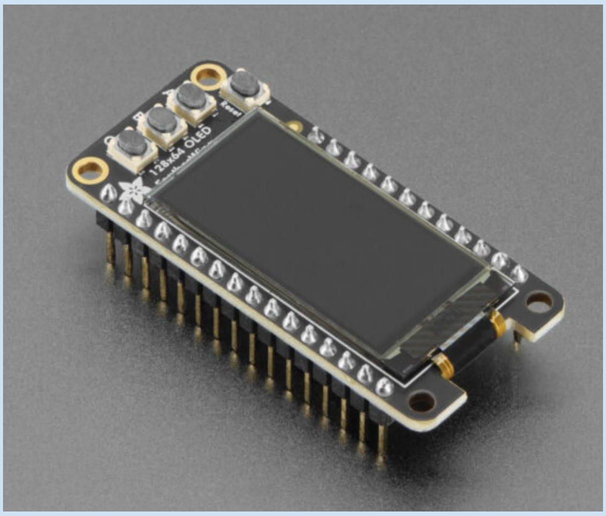
Your ESP32 board is made by a company called [Adafruit](#) in the US and the ESP32 chip it runs on is made by a company called [Espressif](#) in China

Here is a pinout diagram that shows all the connections and what you would label them as in your code. If you have an issue with getting a pin to work I strongly suggest looking at the [page for our board](#) and the pinouts section (paying attention to the orange information bubbles)



Setup stuff - physically setting up the board

Make sure you have soldered pins to your ESP32
Get the teacher to help/inspect your soldering to ensure you have done a good job.



NB: Pins pointing down, screen on top
NB2: This isn't our board, just a clear picture of what we want to do

The teacher will give you a basic 3D printed support for you to screw your ESP32 into so that it is kept secure and slightly more safe than just loose in a cupboard.
You will have the option to remake this, adjust it, or design something different later in the year for another achievement standard.

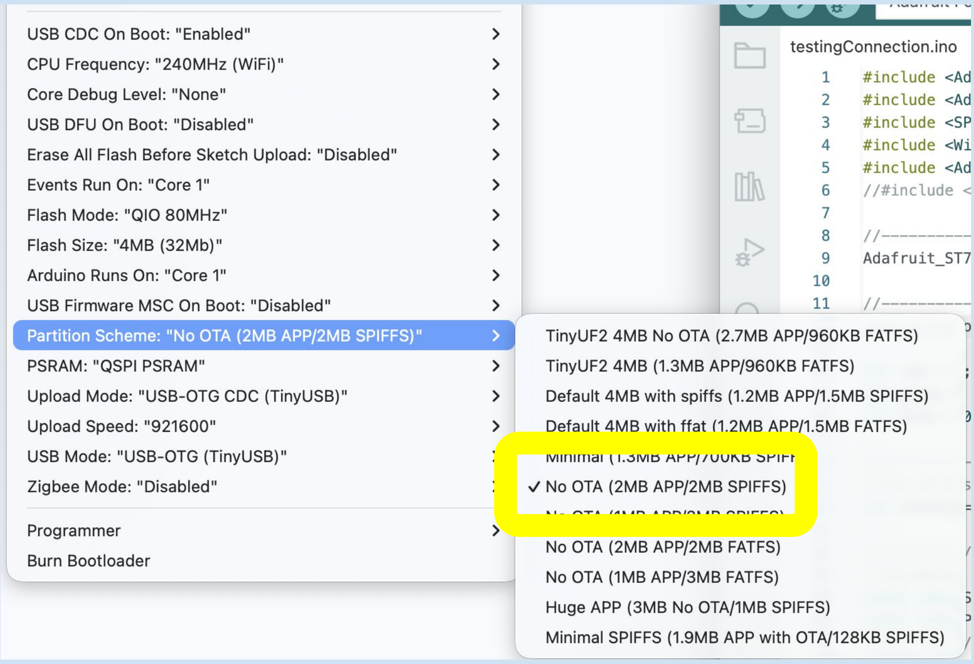
[pic goes here]

Setup stuff - formatting the board

Bring me your board and I will format its memory. This is so everything is set up quickly rather than teaching you something you should only need to do once!

I have used [a site]() to make a custom partitions and then flashed them to your board so that it is set up like this

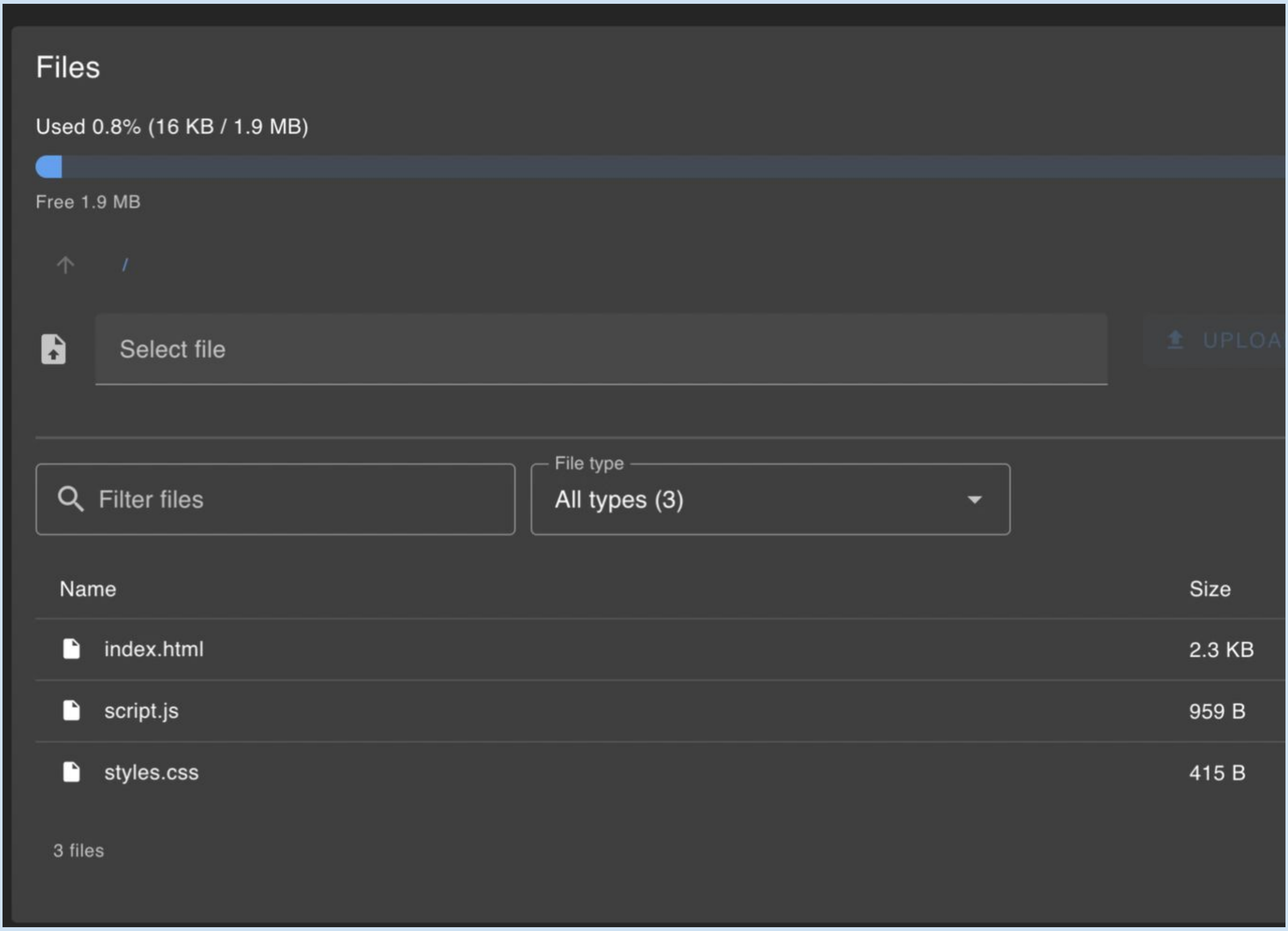
There is 2MB of space for your C++ files in the APP partition and 2MB of space for your HTML, CSS, javascript, and any other files you want to upload in the littleFS partition (although it is named spiffs as the arduino IDE still only wants SPIFFS or FATFS - works fine though)



[Here is]() what you would put it back to if you wanted to reset it to factory

Adding test files

Watch this video for adding the HTML, CSS, JavaScript files using the ESPConnect site

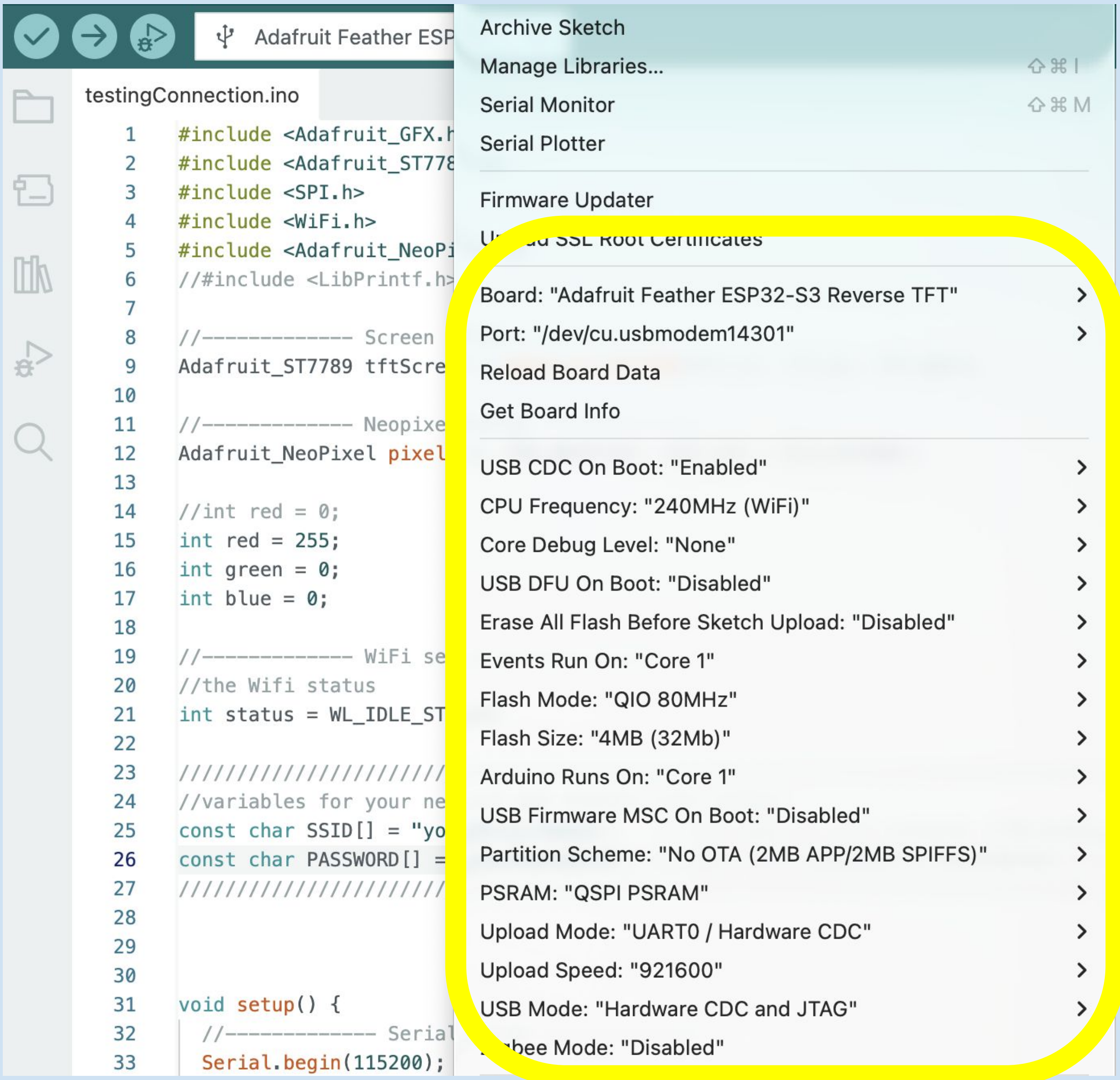


Adding test files

[Watch this video]() for uploading your C++ file to the ESP32

I have used the Arduino IDE to do this but VSCode is also a good IDE Ask me if you need a hand setting it up to upload your code

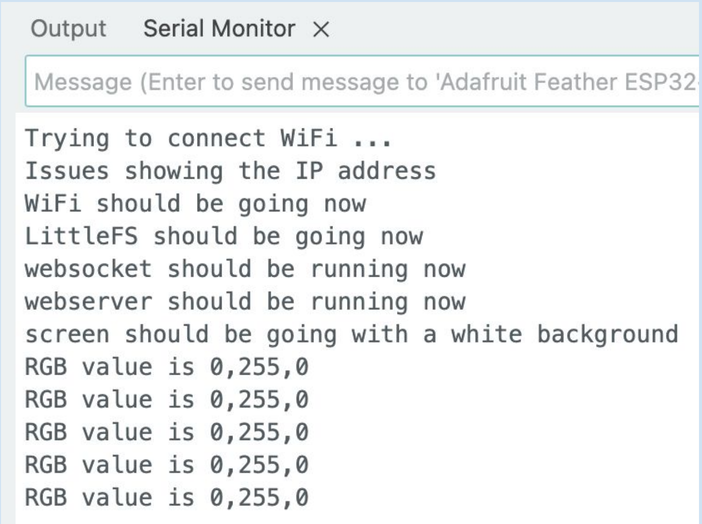
The settings below work well on my apple laptop (and should be fine on any operating system)



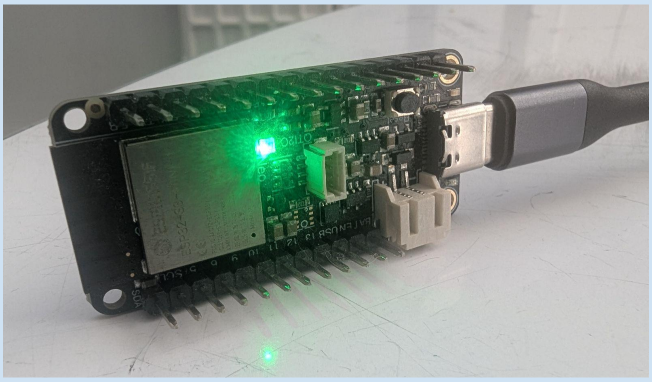
Testing

When the board is first powered on it should

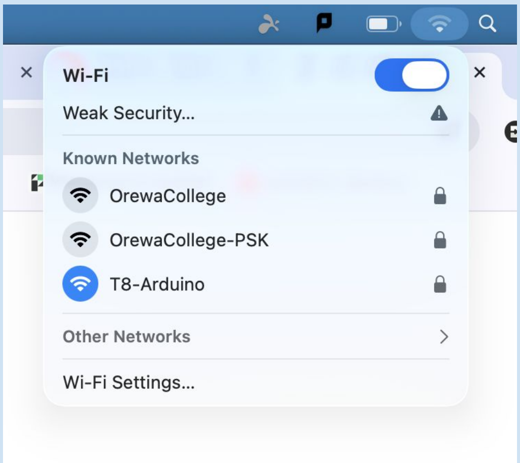
1. Say the following on the serial monitor



2. Say the following on the screen

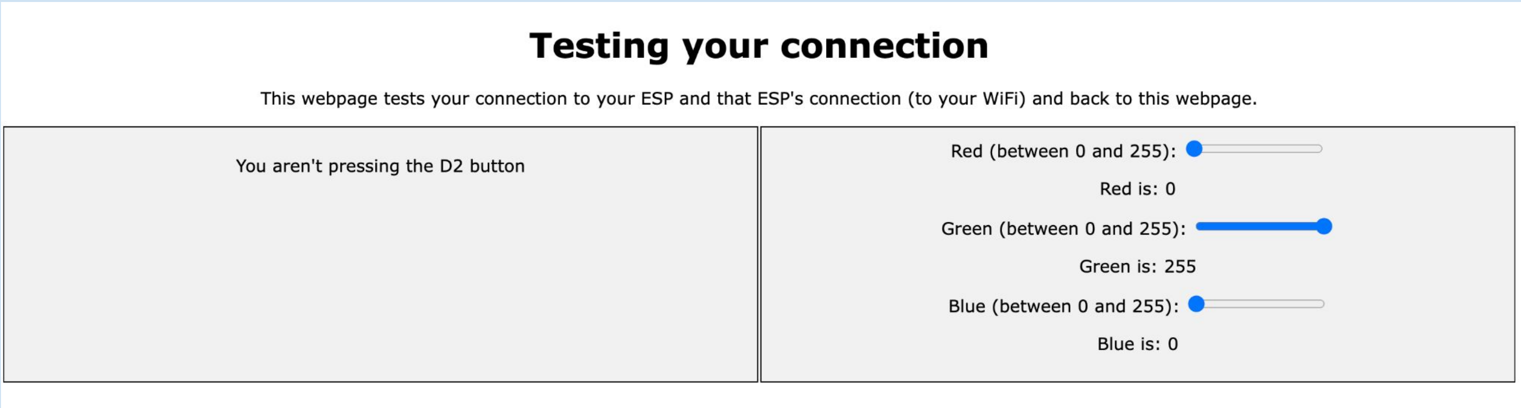


You will need to be on the T8-Arduino wifi to go any further



Testing

3. Once you connect to the ip address it gives show the following on the webpage



Now try adjusting the values on the webpage and see what the neopixel does?

What about pressing the D2 button?